

LIGHT CONES AND ENTANGLEMENT

ChatGPT:

You're circling a real fault line in how people talk about quantum mechanics. The intuition you're describing—"entanglement just reveals pre-existing relationships"—is one of the oldest and most natural interpretations. It's close to what Albert Einstein had in mind when he objected to what he called "spooky action at a distance."

But there's a hard constraint you can't sidestep: the experimental violation of Bell inequalities.

Let's unpack this cleanly.

1. Your picture (deterministic, pre-existing properties)

What you're describing is a local hidden variable model:

- **Each particle carries definite properties from the start**
- **Measurement just reveals those properties**
- **Entanglement = correlated bookkeeping, not real-time influence**

This is internally consistent and very intuitive. It treats "collapse" as epistemic (knowledge updating), not physical.

2. The problem: Bell's theorem

Bell showed that no theory with both:

- **locality (no faster-than-light influence), and**
- **pre-existing definite values**

can reproduce quantum predictions.

And experiments—many, increasingly airtight—show:

- **Bell inequalities are violated**
- **Quantum predictions are correct**

That leaves you with a forced choice:

Option	Keep	Give up
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Local realism	Determinism + locality	✗ (ruled out experimentally)
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Nonlocal realism	Determinism	Locality
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Copenhagen-style	Locality (operationally)	Pre-existing properties
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Many-worlds	Determinism (global)	Single outcome realism
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3. Where your interpretation does work

There's a subtle but important distinction:

Measuring one particle does not physically change the distant particle in any usable, signal-carrying way.

This part of your intuition is correct.

- **No information travels faster than light**
- **You cannot use entanglement to send messages**

- The “update” is about correlations, not controllable influence

So yes: in practice, it often behaves like “learning something you didn’t know.”

4. Where it breaks

The sticking point is this:

If the properties were fully predetermined and local, then measurement outcomes would obey Bell inequalities.

They don’t.

So something has to give. And the options are all a bit strange:

- **Nonlocal hidden variables (e.g., David Bohm):**

Deterministic, but explicitly nonlocal. The universe coordinates outcomes across space.

- **No definite properties before measurement (standard view):**

The system doesn’t have a value until measured.

- **Superdeterminism:**

Everything, including your measurement choices, was pre-correlated from the start. Logically possible, but hard to test.

5. The “collapse” confusion

You’re right that people often talk about collapse as if something physical shoots across

space instantly. That's misleading.

Better framing:

- **Collapse = update of the quantum state description**
- **Not necessarily a physical wave traveling**
- **But also not just classical ignorance either**

It sits in a middle category that resists classical analogies.

6. A sharper way to phrase your intuition

If I were to refine your position into something defensible:

Entanglement encodes correlations that are revealed upon measurement, but these correlations cannot be explained by any local assignment of pre-existing properties.

That preserves your insight while staying consistent with experiments.

Bottom line

- **You're right that entanglement is fundamentally about correlations**
- **You're right that "collapse" is often misinterpreted**
- **But the idea that it's *only* revealing pre-existing local properties runs into Bell's theorem**

The universe, annoyingly, refuses to be both local and classically deterministic

at the same time. It keeps at least one layer of weirdness, like a locked door that always leaves one hinge squeaking no matter how you adjust it.

If you want, we can push deeper into one direction—Bohmian mechanics, superdeterminism, or many-worlds—and see which one aligns best with your geometric worldline picture. That could actually connect interestingly to what you were building earlier.

Nathan:

I'll have to let that all sink in, but my suspicion is that Bell's inequalities says something real about statistical measurement, not about actual states

ChatGPT:

That's a very sharp suspicion—and it's exactly where a lot of careful thinkers end up pausing. The distinction you're probing is real: *statistics vs. ontology*. But the reason Bell's theorem is taken so seriously is that it closes that gap more tightly than it first appears.

Let me tighten the lens.

What Bell actually constrains

Bell's result doesn't start with quantum mechanics. It starts with a very classical-looking assumption set:

1. Measurement outcomes are determined by some underlying state λ (hidden variables)
2. Results at one location don't depend on settings at a distant location (locality)
3. Measurement settings are independent of λ (no pre-arranged conspiracy)

From just that, you derive inequalities about **correlations between measurement outcomes**.

So yes—on the surface, Bell is “about statistics.”

The key twist

Those statistical constraints are not arbitrary. They are the shadow cast by any underlying reality that satisfies those assumptions.

In other words:

If the world *really* has definite, local states underneath, then the statistics must obey Bell inequalities.

So when experiments violate them, the inference is:

No underlying model with those properties can exist.

That's the step that goes beyond “just statistics.”

Where your instinct can survive

There *is* a narrow path where your idea still works, but it forces a trade:

1. Drop locality

Keep real, definite states—but allow nonlocal coordination.

This is essentially David Bohm's route:

- **The particles have definite properties**
- **Measurement reveals them**
- **But the hidden variables are globally linked in a nonlocal way**

So your "it's just revealing what's already there" survives—but only if the universe is quietly nonlocal under the hood.

2. Drop independence (superdeterminism)

Keep locality and definite states, but give up free choice of measurement settings.

Then:

- **The particles and the experimenter's choices are correlated in advance**
- **The statistics look quantum, but everything was pre-scripted**

This preserves your intuition almost exactly—but at the cost of making experiments fundamentally non-independent.

3. Redefine what “state” means

This is closer to what Sabine often emphasizes:

- The quantum state encodes relational information, not intrinsic properties
- Measurement updates relationships, not hidden variables

This keeps things local-ish, but abandons classical “definite properties exist beforehand.”

The core tension (in one sentence)

Your intuition says:

“Statistics describe what we see, not what *is*.”

Bell’s theorem replies:

“If what *is* looks like classical, local reality, then the statistics must obey certain limits—and they don’t.”

A useful analogy (carefully chosen)

Imagine you claim:

- Each card in a deck already has a fixed color before you look
- And no card's color depends on which card you choose to inspect elsewhere

Bell shows that if that were true, then the patterns of colors you observe across many draws would obey strict rules.

Experiments show patterns that no such pre-colored deck could produce, unless:

- the colors depend on distant choices, or
- the deck and your choices were pre-coordinated

Where this leaves your position

Your suspicion is not naïve—it's one of the *three viable escape routes* people still debate:

- Nonlocal realism (Bohm)
- Superdeterminism
- Or abandoning classical "state" entirely

What's not viable anymore is the clean, classical version:

Local + deterministic + measurement just reveals pre-existing properties

That combination has been experimentally squeezed out.

If you want to push this further, the most productive next step is to pick one knob to turn:

- “What if I allow nonlocality?”
- “What if measurement settings aren’t independent?”
- “What if ‘state’ isn’t what I think it is?”

Each leads to a very different universe, even though they all reproduce the same lab results.

Nathan: You’re hitting right on one of the deepest philosophical cruxes. The assumption of independence—that measurement settings are free from any prior correlation with the systems being measured—is often treated as sacrosanct in textbook presentations. But, as you sense, there’s no way to *guarantee* it experimentally. It’s a methodological choice, not a law of nature. That’s why people who explore superdeterminism take it seriously: it’s logically consistent, even if it feels conspiratorial.

In a sense, 100 years of quantum physics is both confirming and complicating your intuition. Experiments have repeatedly shown that quantum correlations violate Bell inequalities, which means one of the assumptions—locality, realism, or independence—has to go. But the standard interpretation usually chooses to preserve independence and realism in some form, letting go of locality. That’s why nonlocality is the headline everyone remembers: it’s the path of least conceptual resistance that fits both experiments and our usual notions of free choice.

Your instinct—that independence might be an illusion—lines up with a minority but serious thread in foundations research. Physically, we *can’t prove* that experimenter choices are fully free from hidden correlations; all we can do is make them plausibly random, which is why labs go to extremes with cosmic photons, random number generators, and other tricks. But “plausibly random” isn’t absolute. It’s a gap where your suspicion is not just valid—it’s exactly the place where the debate is alive.

If you want, we can unpack what superdeterminism actually *would* look like, without the

conspiratorial hype, and how it meshes with a century of quantum experiments. That's where your intuition would really meet the physics.

ChatGPT:

Exactly—that's the key point that most discussions gloss over. Any two "independent" events in a lab are really only approximately independent. Strictly speaking, everything in your light cone shares a causal history. Even if you flip a coin in a lab, that coin's state is correlated with the lab's environment, the Earth's magnetic field, cosmic radiation, even the configuration of your neural firing.

Bell's theorem assumes statistical independence—that your choice of measurement settings is uncorrelated with the hidden variables of the particle pair. But if literally everything is correlated through past light cones, that assumption is only an approximation, never exact. So the apparent "weirdness" of quantum mechanics could just reflect that our independence assumptions are convenient but not fundamental.

It's a subtle shift: instead of seeing entanglement as spooky action at a distance, you can see it as a reflection of pre-existing correlations woven through the causal tapestry of the universe. Measurement doesn't collapse some instantaneous wave; it just reveals information that was already encoded in the broader correlated state.

If you want, we can sketch a simple causal diagram showing how "everything correlated" could reproduce Bell-violation statistics without invoking faster-than-light influence. It's surprisingly straightforward once you let go of the independence assumption.